

VrindaaCorp Services — Database ERD & Complete Data Dictionary

Confidential & Proprietary • Engineering Architecture Reference

System: VrindaaCorp Enterprise CRM & Autonomous Outreach Infrastructure

Database Engine: PostgreSQL 16.x • ORM: Prisma 5.20.0 • Schema: public

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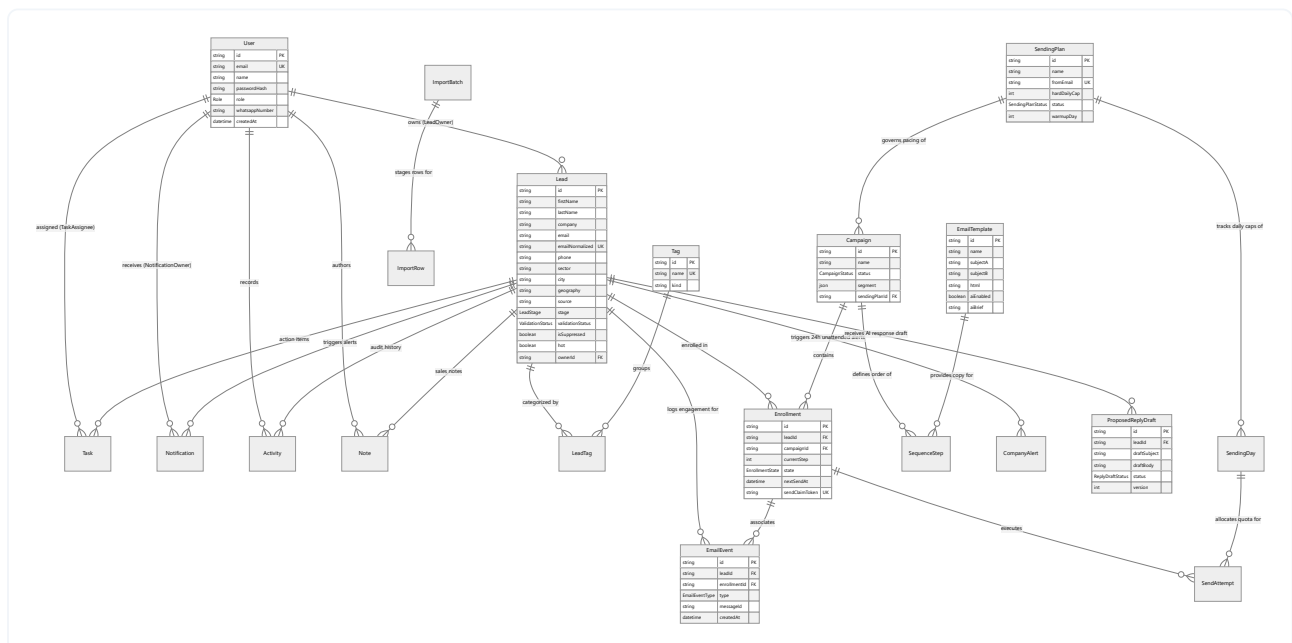
1. Architectural Overview & Design Principles

The VrindaaCorp CRM database is engineered to handle high-concurrency lead lifecycle tracking, automated campaign sequence execution, real-time email engagement metrics, and AI-driven conversational approvals.

Key Database Characteristics:

- **ACID Compliance:** Guaranteed transactional safety across batch email enrollments, deduplication runs, and status mutations.
- **Relational Integrity:** Explicit foreign key constraints with appropriate `ON DELETE CASCADE` or `ON DELETE SET NULL` cascades to prevent orphaned child records.
- **Normalized Query Optimization:** Composite B-tree indexes applied on high-cardinality filters (`emailNormalized`, `stage`, `sector`, `geography`, `ownerId`, and time-series send timestamps).
- **Scalability:** 24 models and 13 enumerated types supporting millions of lead interaction events with zero schema drift.

2. Complete Entity-Relationship Diagram (ERD)



3. Enumerated Types Reference (Enums)

Enum Identifier	Declared Values	Functional Purpose in System
Role	ADMIN, OWNER, AGENT	Role-based access control governing CRM navigation, user management, and campaign controls.
LeadStage	NEW, CONTACTED, REPLIED, QUALIFIED, PROPOSAL_SENT, WON, LOST	7-stage sales pipeline tracking a prospect from cold outreach to signed deal.
ValidationStatus	UNKNOWN, VALID, INVALID, RISKY, DISPOSABLE, CATCH_ALL	Result of the 6-layer in-house email verification engine.
ContractStatus	UNKNOWN, NONE, ACTIVE	Incumbent facility vendor contract tracking.
ImportStatus	PENDING, MAPPED, PROCESSING, COMPLETED, FAILED	Staging status for bulk CSV/Excel lead imports.
CampaignStatus	DRAFT, ACTIVE, PAUSED, COMPLETED	Execution state of an outreach campaign.
SendingPlanStatus	DRAFT, ACTIVE, PAUSED, COMPLETED	Operational status of the domain warmup plan.
SendAttemptStatus	PENDING, SENT, FAILED, UNKNOWN	Result of an individual email transmission attempt.
EnrollmentState	ACTIVE, PAUSED, COMPLETED	Status of a specific lead progressing through sequence steps.
EmailEventType	SENT, DELIVERED, OPENED, CLICKED, REPLIED, BOUNCED, COMPLAINED, UNSUBSCRIBED	Audit log for tracking pixels, clicks, and mailbox replies.
SuppressionReason	HARD_BOUNCE, COMPLAINT, UNSUBSCRIBE, MANUAL	Reason an email address is permanently barred from outreach.
NotificationState	SENT, ACKNOWLEDGED, ESCALATED	State of an owner notification alert.
ReplyDraftStatus	PENDING_APPROVAL, APPROVED, REVISED, SENT, REJECTED	WhatsApp approval status of an AI-generated revert draft.

4. Complete Data Dictionary (All 24 Models)

4.1 User (Authentication & Team Members)

Stores credentials, access roles, and notification phone numbers for sales and administrative staff.

Column	Type	Nullable	Default	Constraints	Description
id	TEXT	No	cuid()	PK	Unique record identifier
email	TEXT	No	—	UNIQUE	Corporate email address (login credential)
name	TEXT	Yes	—	—	Full display name
passwordHash	TEXT	No	—	—	Bcrypt hashed password (salt cost: 12)
role	Role	No	AGENT	—	Access role: ADMIN , OWNER , AGENT
whatsappNumber	TEXT	Yes	—	—	E.164 phone number for WhatsApp alerts (+91 ...)
createdAt	TIMESTAMP	No	now()	—	Account creation timestamp
updatedAt	TIMESTAMP	No	—	—	Last account update timestamp

4.2 Lead (Primary Prospect Data Store)

The central entity representing corporate prospects across target sectors.

Column	Type	Nullable	Default	Constraints	Description
<code>id</code>	TEXT	No	<code>cuid()</code>	PK	Unique lead identifier
<code>firstName</code>	TEXT	Yes	—	—	Contact first name
<code>lastName</code>	TEXT	Yes	—	—	Contact last name
<code>company</code>	TEXT	Yes	—	—	Company / Enterprise name
<code>email</code>	TEXT	No	—	—	Contact email address
<code>emailNormalized</code>	TEXT	No	—	INDEX	Lowercased/trimmed email for dedup & suppression
<code>phone</code>	TEXT	Yes	—	—	Phone number formatted in E.164 standard
<code>sector</code>	TEXT	Yes	—	INDEX	Target sector: <code>Corporate</code> , <code>Healthcare</code> , <code>Industrial</code> , etc.
<code>city</code>	TEXT	Yes	—	—	Normalized city (e.g., <code>Gurgaon</code> , <code>Noida</code> , <code>Lucknow</code>)
<code>geography</code>	TEXT	Yes	—	INDEX	Regional hub: <code>NCR</code> , <code>UP</code> , <code>pan-India</code>
<code>source</code>	TEXT	Yes	—	—	Source channel: <code>website_form</code> , <code>meta_ads</code> , <code>import</code>
<code>sourceDetail</code>	TEXT	Yes	—	—	Form ID, campaign ID, or CSV filename
<code>stage</code>	LeadStage	No	NEW	INDEX	Sales pipeline stage
<code>validationStatus</code>	ValidationStatus	No	UNKNOWN	—	Result of 6-layer cleansing engine
<code>validationReason</code>	TEXT	Yes	—	—	Human-readable explanation (e.g., <code>Disposable domain</code>)
<code>validationCheckedAt</code>	TIMESTAMP	Yes	—	—	Timestamp of last validation run
<code>smtpCheckedAt</code>	TIMESTAMP	Yes	—	INDEX	Timestamp of live DNS/SMTP MX probe
<code>isSuppressed</code>	BOOLEAN	No	false	—	True if unsubscribed, bounced, or suppressed
<code>hot</code>	BOOLEAN	No	false	—	True if prospect replied or engaged multiple times
<code>ownerId</code>	TEXT	Yes	—	FK → User(id)	Assigned sales representative
<code>contractStatus</code>	ContractStatus	No	UNKNOWN	—	Incumbent vendor contract status
<code>incumbentVendor</code>	TEXT	Yes	—	—	Name of current facility vendor if known
<code>contractExpiry</code>	TIMESTAMP	Yes	—	INDEX	Expiration date of incumbent contract

Column	Type	Nullable	Default	Constraints	Description
contractConfidence	TEXT	Yes	—	—	AI confidence level (low , medium , high)
contractSource	TEXT	Yes	—	—	Source: ai , manual , import
contractCheckedAt	TIMESTAMP	Yes	—	INDEX	Timestamp of contract intelligence check
contractConfirmed	BOOLEAN	No	false	—	Verified by human sales agent
contractReminderSentAt	TIMESTAMP	Yes	—	—	30-day renewal reminder timestamp
createdAt	TIMESTAMP	No	now()	—	Timestamp of creation
updatedAt	TIMESTAMP	No	—	—	Timestamp of last modification

4.3 DomainReputation (DNS & Catch-All Cache)

Caches domain-level MX and catch-all results to prevent redundant DNS queries across identical company domains.

Column	Type	Nullable	Default	Constraints	Description
domain	TEXT	No	—	PK	Domain name (e.g., infosys.com)
isCatchAll	BOOLEAN	No	false	—	True if domain accepts all inbound mailboxes
mxHost	TEXT	Yes	—	—	Primary Mail Exchange (MX) hostname
lastCheckedAt	TIMESTAMP	No	now()	—	Cache timestamp

4.4 Tag & LeadTag (Flexible Multi-Tagging)

Manages many-to-many tag relationships (e.g., sectors, priority tiers, campaign buckets).

Model	Column	Type	Constraints	Description
Tag	id	TEXT	PK	Unique tag ID
	name	TEXT	UNIQUE	Tag name (e.g., Healthcare , NCR , Hot-Lead)
	kind	TEXT	Default: general	Category: sector , geography , priority , campaign
LeadTag	leadId	TEXT	PK, FK → Lead(id)	Associated lead (Cascade on delete)
	tagId	TEXT	PK, FK → Tag(id)	Associated tag (Cascade on delete)

4.5 ImportBatch & ImportRow (Spreadsheet Staging)

Provides resilient 2-phase staging for CSV/Excel file uploads.

Model	Column	Type	Constraints	Description
ImportBatch	id	TEXT	PK	Unique batch identifier
	filename	TEXT	—	Uploaded spreadsheet name
	status	ImportStatus	Default: PENDING	PENDING, MAPPED, PROCESSING, COMPLETED
	columnMapping	JSONB	—	Column map { email: "Work Email", company: "Org" }
	totalRows	INT	Default: 0	Total records in file
	importedRows	INT	Default: 0	Successfully validated & imported records
	duplicateRows	INT	Default: 0	Duplicate records skipped
ImportRow	invalidRows	INT	Default: 0	Malformed records rejected
	id	TEXT	PK	Unique row identifier
	batchId	TEXT	FK → ImportBatch(id)	Parent batch ID
	raw	JSONB	—	Raw JSON row before transformation
	status	TEXT	Default: pending	pending, imported, duplicate, invalid
	note	TEXT	—	Processing note or validation failure reason

4.6 SendingPlan & SendingDay (Outreach Warmup Engine)

Enforces mathematical domain warmup curves, daily caps, and automated safety stops.

Model	Column	Type	Constraints	Description
SendingPlan	id	TEXT	PK	Unique sending plan identifier
	name	TEXT	—	Plan name (e.g., <code>Sales Primary Warmup</code>)
	fromEmail	TEXT	UNIQUE	Outbound sender address (<code>sales@vrindaacorp.com</code>)
	fromDomain	TEXT	INDEX	Outbound domain (<code>vrindaacorp.com</code>)
	timezone	TEXT	Default: <code>Asia/Kolkata</code>	Operational business timezone
	sendWindowStart	TEXT	Default: <code>09:00</code>	Start of daily dispatch window (IST)
	sendWindowEnd	TEXT	Default: <code>18:00</code>	End of daily dispatch window (IST)
	hardDailyCap	INT	Default: <code>1000</code>	Maximum absolute ceiling
	status	SendingPlanStatus	Default: <code>DRAFT</code>	<code>DRAFT</code> , <code>ACTIVE</code> , <code>PAUSED</code> , <code>COMPLETED</code>
	warmupDay	INT	Default: <code>1</code>	Current day in the warmup schedule
	schedule	JSONB	—	Array of daily allowed limits by warmup day
	bounceStopRate	FLOAT	Default: <code>0.05</code>	Auto-pause threshold (5% bounce rate)
SendingDay	id	TEXT	PK	Unique record identifier
	planId	TEXT	FK → <code>SendingPlan(id)</code>	Associated sending plan
	localDate	DATE	INDEX	Calendar day in target timezone
	allowed	INT	—	Max emails permitted today
	reserved	INT	Default: <code>0</code>	Emails claimed by worker
	sent	INT	Default: <code>0</code>	Emails successfully dispatched
	failed	INT	Default: <code>0</code>	Email dispatch errors
	bounced	INT	Default: <code>0</code>	Bounces recorded
	complained	INT	Default: <code>0</code>	Spam complaints recorded

4.7 Campaign, EmailTemplate & SequenceStep (Sequence Builder)

Manages multi-step automated email workflows with dynamic sector copy injection.

Model	Column	Type	Constraints	Description
Campaign	id	TEXT	PK	Unique campaign identifier
	name	TEXT	—	Campaign title
	status	CampaignStatus	Default: DRAFT	DRAFT , ACTIVE , PAUSED , COMPLETED
	segment	JSONB	—	Filter JSON { sector, geography, tags }
	sendingPlanId	TEXT	FK → SendingPlan(id)	Bound domain warmup plan
EmailTemplate	id	TEXT	PK	Unique template identifier
	name	TEXT	—	Template display name
	subjectA	TEXT	—	Primary email subject line
	subjectB	TEXT	—	Optional A/B test variant subject line
	html	TEXT	—	HTML body copy with tokens ({{firstName}} , etc.)
	aiEnabled	BOOLEAN	Default: false	Enable per-lead dynamic LLM personalization
	aiBrief	TEXT	—	Specific value proposition instructions for the AI
SequenceStep	id	TEXT	PK	Unique step identifier
	campaignId	TEXT	FK → Campaign(id)	Parent campaign
	templateId	TEXT	FK → EmailTemplate(id)	Associated template copy
	order	INT	—	Step position (0 = Initial, 1 = Follow-up 1)
	delayDays	INT	—	Delay days after preceding step (0 , 3 , 6)

4.8 Enrollment & SendAttempt (Lead Sequence Execution)

Tracks the exact progression of a lead through sequence steps.

Model	Column	Type	Constraints	Description
Enrollment	id	TEXT	PK	Unique enrollment identifier
	leadId	TEXT	FK → Lead(id)	Enrolled prospect
	campaignId	TEXT	FK → Campaign(id)	Active campaign
	currentStep	INT	Default: 0	Index of sequence step to execute
	state	EnrollmentState	Default: ACTIVE	ACTIVE, PAUSED, COMPLETED
	pausedReason	TEXT	—	Reason: replied, unsubscribed, bounced
	nextSendAt	TIMESTAMP	INDEX	Next scheduled dispatch datetime
	sendClaimToken	TEXT	UNIQUE	Distributed atomic lock token preventing double sends
	sendClaimedUntil	TIMESTAMP	INDEX	Lock expiration timestamp (5-minute lease)
SendAttempt	id	TEXT	PK	Unique attempt identifier
	enrollmentId	TEXT	FK → Enrollment(id)	Associated enrollment
	stepOrder	INT	—	Step index attempted
	attemptNumber	INT	—	Retry attempt count
	status	SendAttemptStatus	Default: PENDING	PENDING, SENT, FAILED, UNKNOWN
	providerId	TEXT	—	SMTP messageId or SES provider ID
	error	TEXT	—	Detailed transmission error message if failed

4.9 EmailEvent & Suppression (Tracking & Compliance)

Maintains 100% auditability of deliverability, opens, link clicks, replies, and unsubscribes.

Model	Column	Type	Constraints	Description
EmailEvent	id	TEXT	PK	Unique event identifier
	leadId	TEXT	FK → Lead(id)	Associated prospect
	enrollmentId	TEXT	FK → Enrollment(id)	Associated campaign enrollment
	type	EmailEventType	INDEX	SENT , OPENED , CLICKED , REPLIED , BOUNCED
	messageId	TEXT	INDEX	Correlating message identifier
	subjectVariant	TEXT	—	Variant tracked: "A" or "B"
	metadata	JSONB	—	Request IP, user agent, or clicked URL
	createdAt	TIMESTAMP	INDEX	Event timestamp
Suppression	id	TEXT	PK	Unique suppression ID
	emailNormalized	TEXT	UNIQUE	Suppressed email address
	reason	SuppressionReason	—	HARD_BOUNCE , COMPLAINT , UNSUBSCRIBE
	createdAt	TIMESTAMP	—	Timestamp of suppression

4.10 Activity, Note & Task (CRM Operational Tools)

Provides sales agents with customer timeline history, internal notes, and follow-up tasks.

Model	Primary Columns	Key Foreign Keys	Purpose
Activity	id , type , message , createdAt	leadId → Lead , userId → User	Audit log of stage shifts, calls, and emails
Note	id , body , createdAt	leadId → Lead , userId → User	Internal sales team notes on lead profile
Task	id , title , dueAt , completed	leadId → Lead , assigneeId → User	Action item (e.g., "Send commercial proposal")

4.11 Notification, CompanyAlert & JobRun (Alerts & Supervision)

Drives the 48-hour owner escalation pipeline and background worker health monitoring.

Model	Primary Columns	Key Foreign Keys	Purpose
Notification	id , channels , state , context , acknowledgedAt , escalatedAt	leadId → Lead , ownerId → User	Hot lead notification & 48h owner escalation
CompanyAlert	id , kind , sentAt	leadId → Lead	24h unattended lead reminder to shared corporate inbox
JobRun	id , job , startedAt , finishedAt , ok , detail	—	Heartbeat and diagnostic logs for PM2 worker tasks

4.12 InboundLeadLog & ProposedReplyDraft (AI & Inbound Audit)

Stores external inbound webhook audit trails, persistent Owner Strategic Memory directives, and WhatsApp reply drafts.

Model	Column	Type	Constraints	Description
InboundLeadLog	id	TEXT	PK	Unique record identifier
	channel	TEXT	INDEX	Channel: website_form , meta_ads , owner_strategic_memory
	status	TEXT	—	Ingestion status: received , created , duplicate , active
	payload	JSONB	—	Complete raw request payload / Strategic Memory playbook
	leadId	TEXT	—	Resulting Lead ID if created
	note	TEXT	—	Diagnostic parsing note
ProposedReplyDraft	id	TEXT	PK	Unique draft identifier
	leadId	TEXT	FK → Lead(id)	Associated prospect who replied
	inboundSubject	TEXT	—	Subject line of incoming prospect email
	inboundBody	TEXT	—	Body text of incoming prospect email
	draftSubject	TEXT	—	AI-generated reply subject line
	draftBody	TEXT	—	AI-generated reply body (enforcing Calendly link)
	status	ReplyDraftStatus	Default: PENDING_APPROVAL	PENDING_APPROVAL , APPROVED , REVISED , SENT
	agentPhone	TEXT	INDEX	WhatsApp recipient phone number
	version	INT	Default: 1	Revision version number
	revisionNotes	TEXT	—	Owner feedback provided via WhatsApp chat

End of Database ERD & Data Dictionary.